

simplified and reusable curve fitting

SplineCloud : Open Platform for Formalized Knowledge Exchange

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by AlinaNeh ★ 2 684 (1) 1827 (2)

Airfoil database containing the main aerodynamic characteristics of symmetrical airfoils developed by different organizations.

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	Alpha	Cl	Cd	Cdp	Cm	Top_Xtr	Bot_Xtr
0	-10.50	-0.7131	0.07555	0.07014	-0.0405	1.0000	0.0995
1	-10.25	-0.7225	0.07035	0.06491	-0.0414	1.0000	0.0980
2	-10.00	-0.7491	0.06473	0.05918	-0.0416	1.0000	0.0962
3	-9.75	-0.7792	0.05833	0.05251	-0.0411	1.0000	0.0941

Files

NACA 63-215

NACA 63-215 (cl_aoa).png

NACA 63-218

NACA 63-218 (cl_aoa).png

NACA 63-221

NACA 63-415

NACA 63-418

NACA 63-421

NACA 64-415

NACA 64-421

Display: Open

NACA 63-215 / NACA 63-215 (cl_aoa) / NACA 63-215 Cl(A...

Y axis data: Cl

X axis data: Angle of Attack (deg)

Curves

Spline Curve 1

Type: Smoothing

Smoothness: 0.05

Order: 2

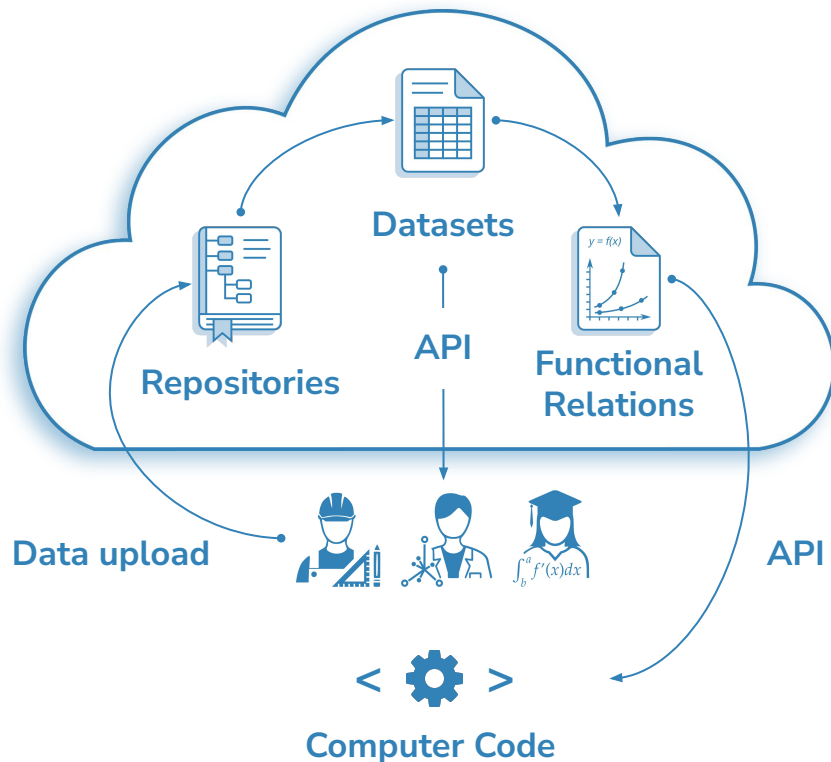
Open and free to use 

Interactive curve fitting 

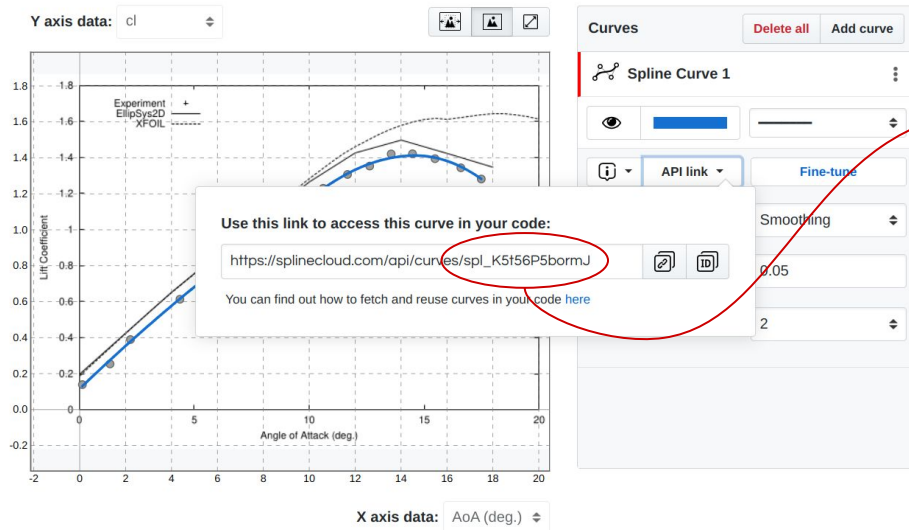
API for data and functions 

Few Steps to Reusability of Regression Models

1. **Create** topical **repository** (add short description, topics and wiki to make it findable)
2. **Upload data** (spreadsheets, tables, catalogues, graphs)
3. **Define/Create datasets** (use plot digitizing tool to extract data from graphs)
4. **Build functional relations** (use advanced spline fitting tool to find relations in data)
5. **Reuse** datasets and curves in code over API



Access and Reuse Spline Curves in Code

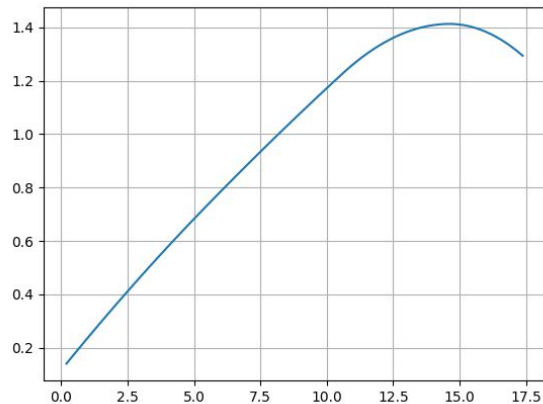


```
from splinecloud_scipy import load_spline

spline = load_spline('spl_K5t56P5bormJ')

X = np.linspace(0, 20, 200)
Y = spline.eval(X)

plt.plot(X, Y)
plt.show()
```

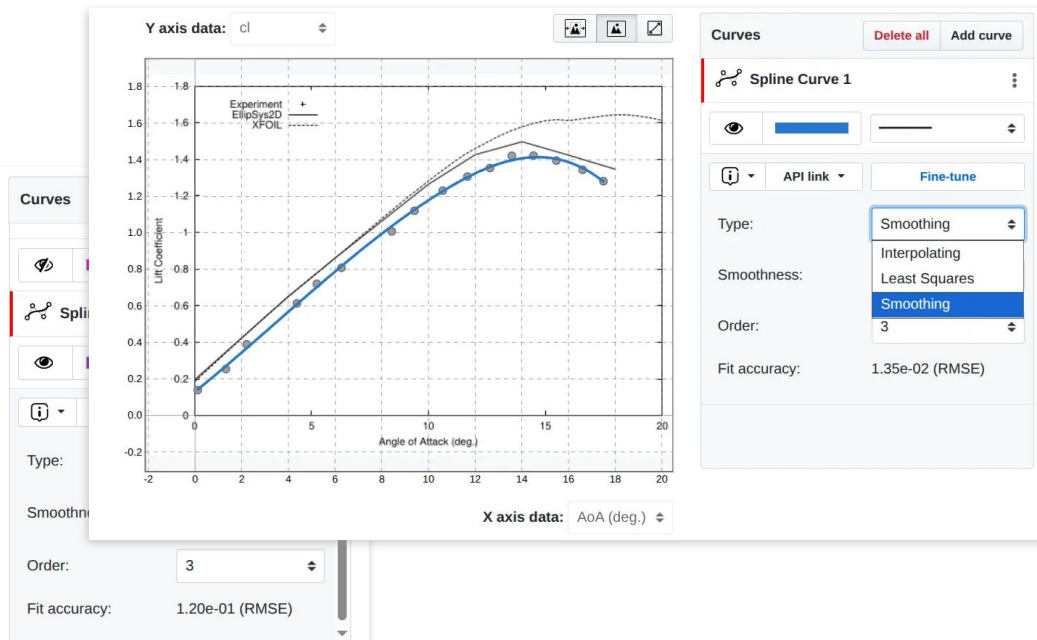
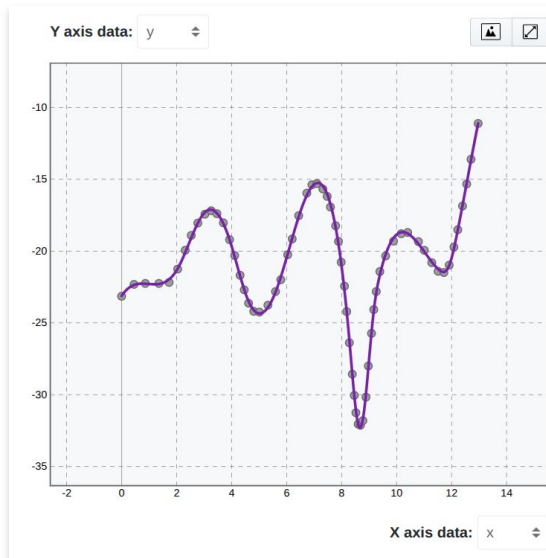


Install client for Python

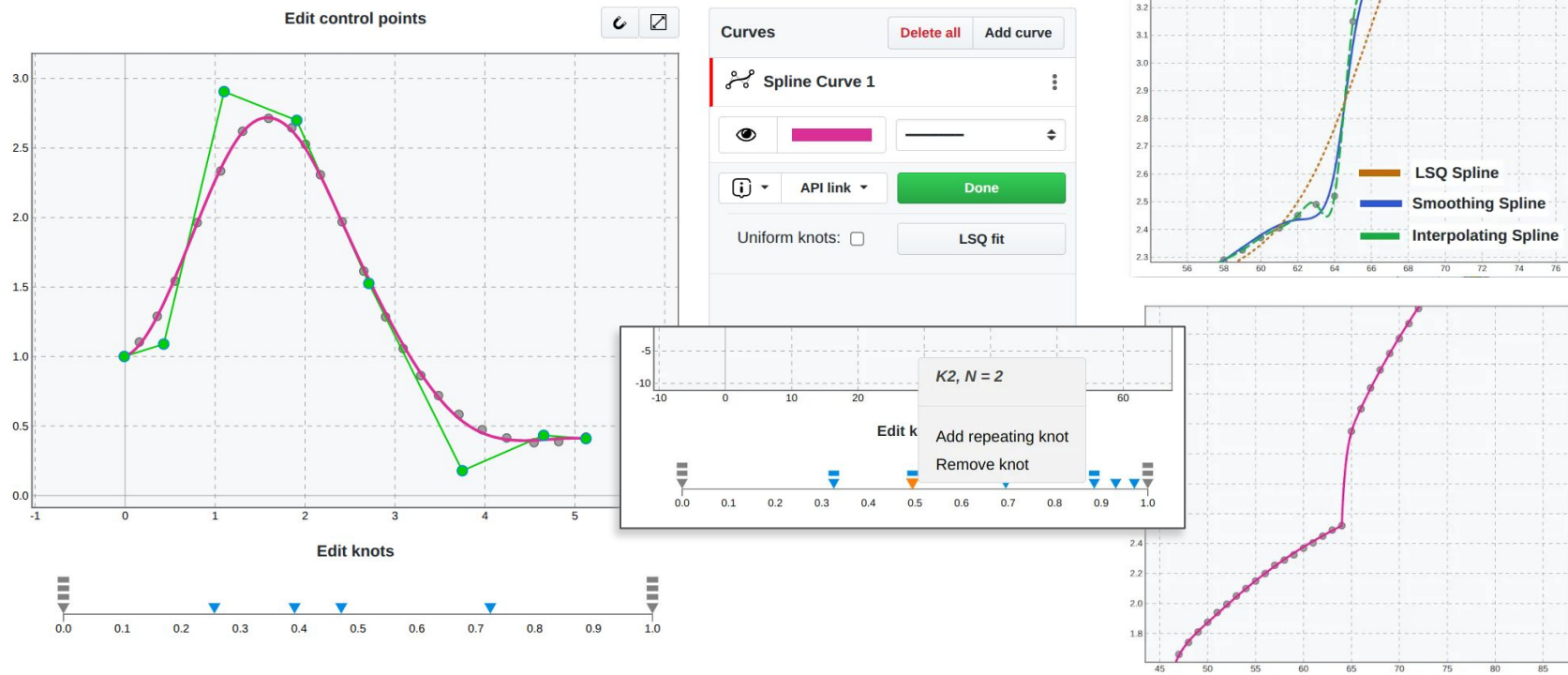
```
pip install splinecloud-scipy
```

Features

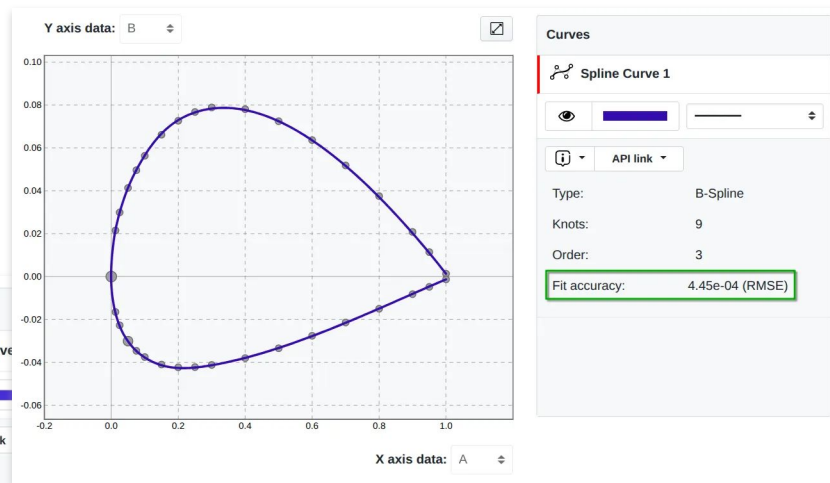
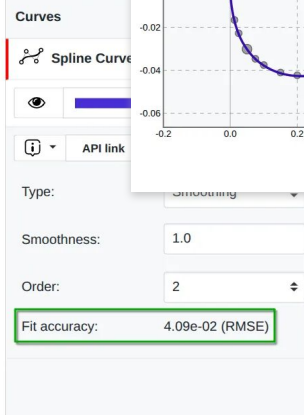
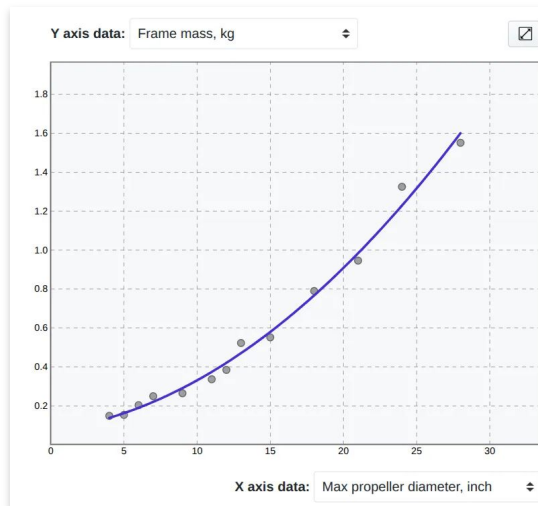
Automatic Spline Fitting Algorithms: SciPy Methods



Interactie Fine Tuning Mode to Edit Cure Shapes Using Handlers



Estimating Fit Accuracy

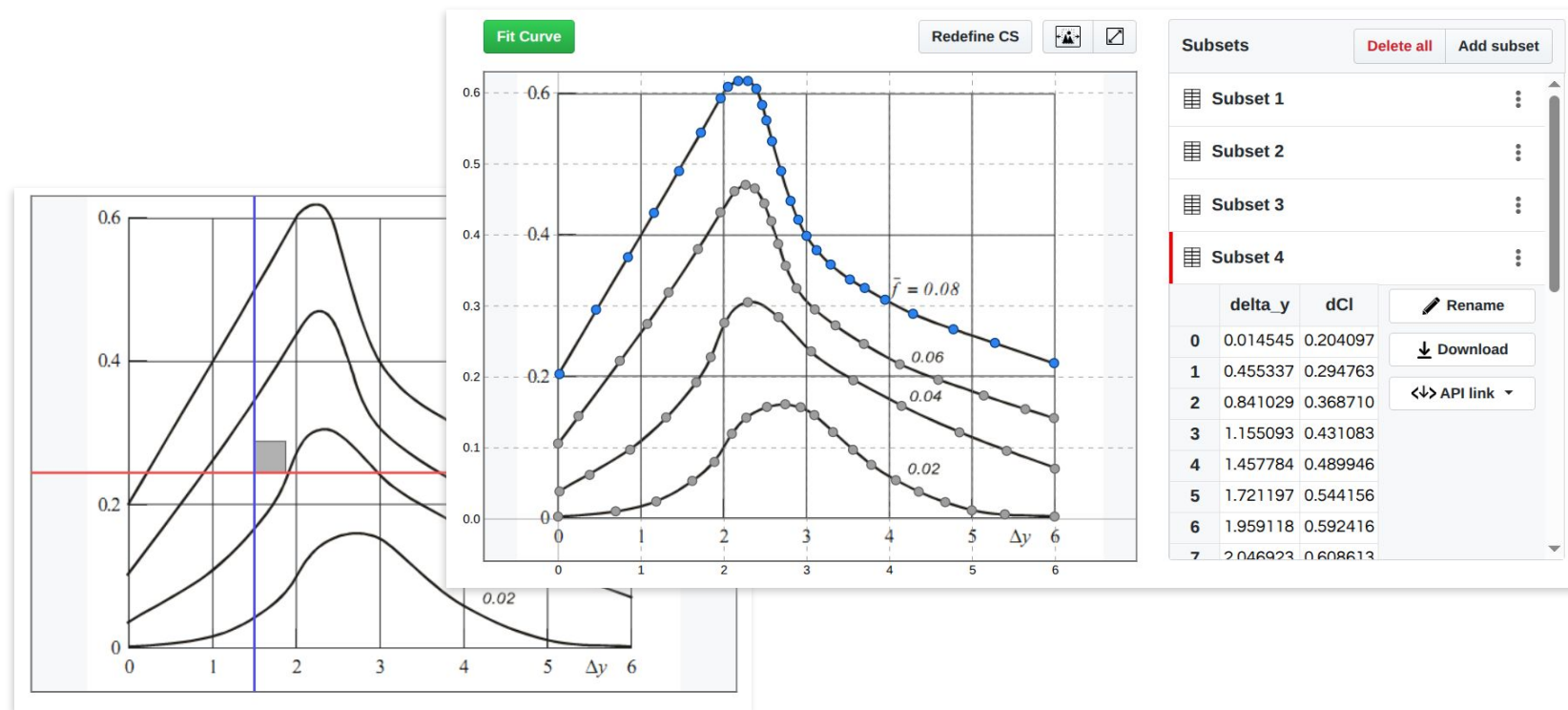


$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - S(x_i)|$$

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - S(x_i))^2$$

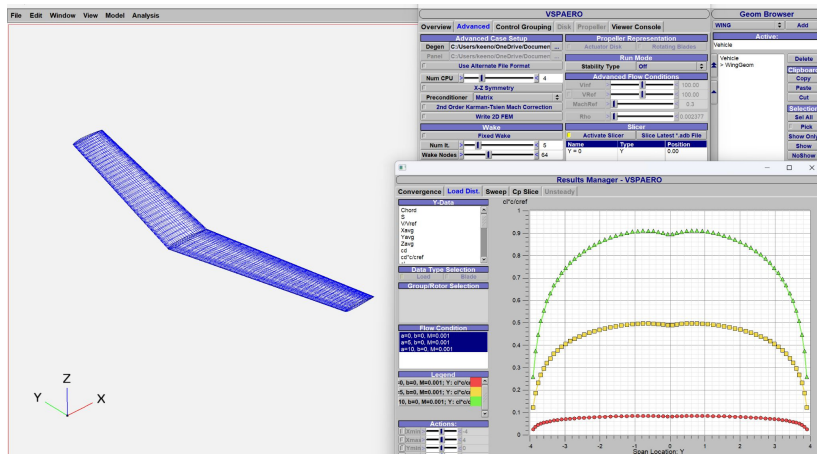
$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - S(x_i))^2}$$

Plot Digitizer



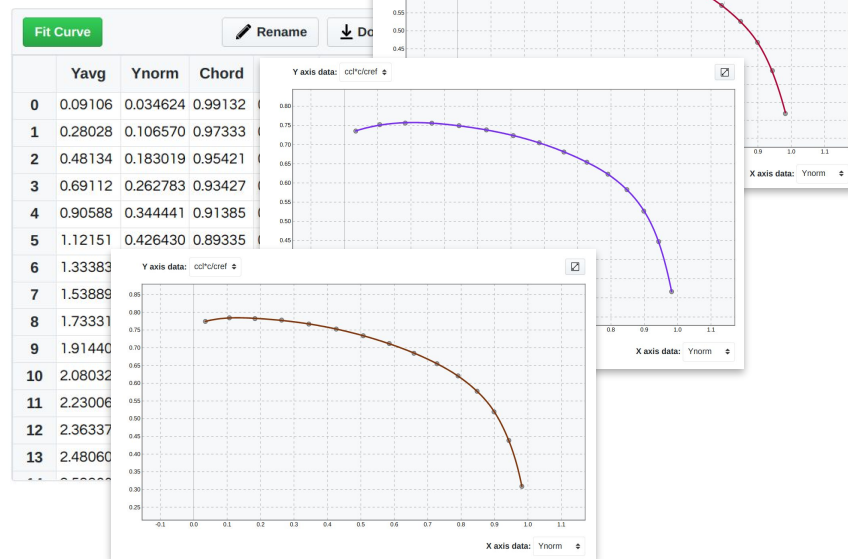
Reusability Examples

Example: Building Reusable Functional Relations Based on Numerical Analysis



AR=6 AoA=0

NACA 1412 / [naca1412_taper_0.75_gamma_1](#)

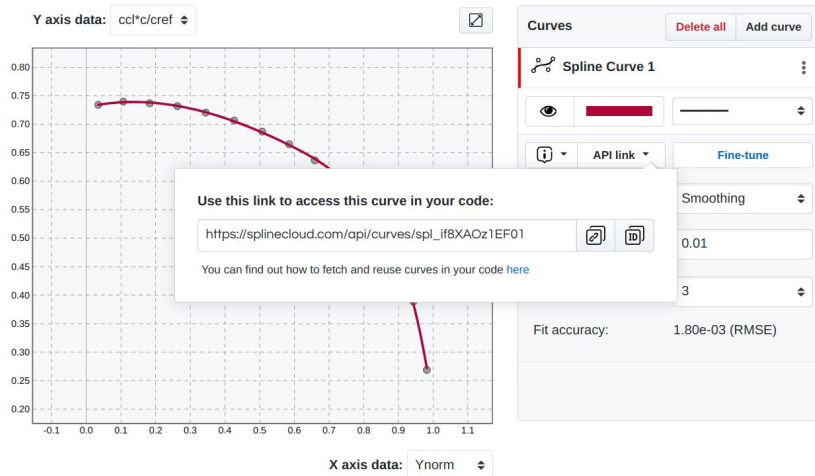


Numerical Analysis in OpenVSP

Reusing Collection of Curves in Code

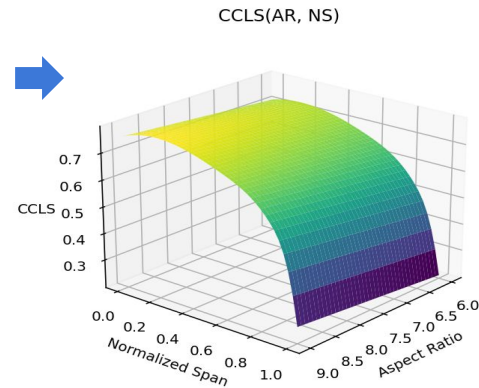
CCI*c/cref by span (Gamma=12 AR=6 AoA=0)

NACA 1412 / [naca1412_taper_0.75_gamma_12.xlsx](#) / AR=6 AoA=0



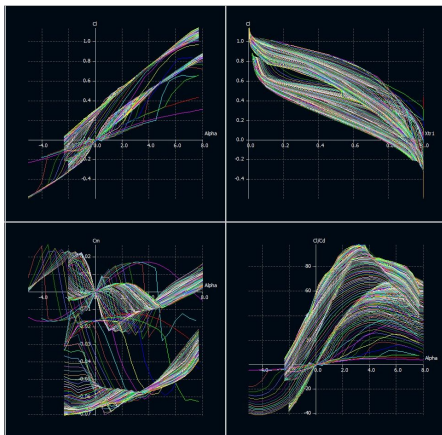
```
ccls_data = {
  "gamma=12": {
    "alpha=0": {
      6: "spl_if8XAOz1EF01",
      7: "spl_QyMqI7K4dyEe",
      8: "spl_qk3ZnLPcdim2",
      9: "spl_rPgCZn0E1Bc4"
    },
    "alpha=5": {
      6: "spl_JvesBTd0ISv5",
      7: "spl_V0SHgq0jfVq1",
      8: "spl_3bGg9cNVK9kj",
      9: "spl_lyEFFznnRiS3"
    },
    "alpha=10": {
      6: "spl_VLW7VrFbfws2",
      7: "spl_aeTTLUK15Byg",
      8: "spl_ORbrtErrHW1h",
      9: "spl_OutD1lpXbXMw"
    }
  },
  "gamma=15": {
    "alpha=0": {
      6: "spl_Tdh4VJ5dNXIi",
      7: "spl_SX0MQ3ZNsqn",
      8: "spl_5aTXb1NVrGYm",
      9: "spl_Kr8AxTPgoyLd"
    },
    "alpha=5": {
      6: "spl_AtW6Ln1Clg7z",
      7: "spl_ONuF07XzDYza",
      8: "spl_BMyNr3Wynw1W",
      9: "spl_Tc7Wfoh4tSPb"
    },
    "alpha=10": {
      6: "spl_TJH9zbnkH1Sh",
      7: "spl_eu8e12uitQKn",
      8: "spl_2sca03Ca6i1W",
      9: "spl_toFY32FcHmtJ"
    }
  }
}
```

Wing lift efficiency coefficient

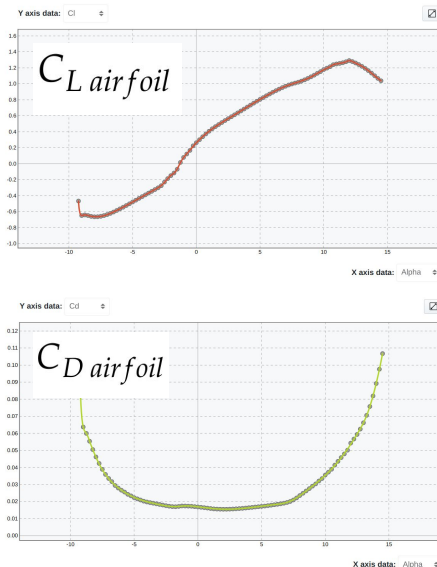


Example: Building a Library of Airfoil Performance Curves

XFOIL



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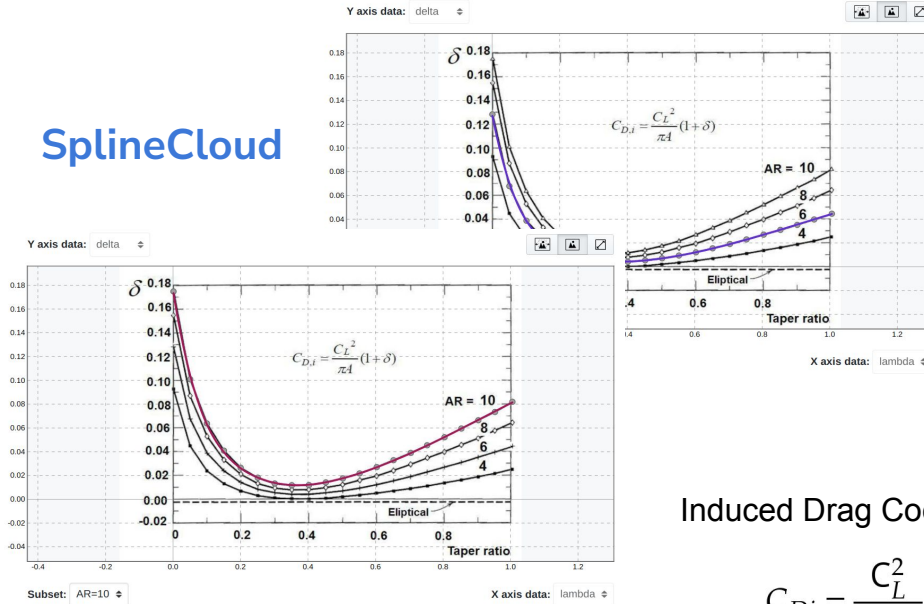
Python

```
{
  "NACA 4412": {
    "name": "NACA 4412 (naca4412-il)",
    "curves": {
      "CL_1000000": "spl_5MJnqWp8tFjk",
      "CL_500000": "spl_cTj0LM7UbRHv",
      "CL_200000": "spl_ajJkUYMsRf2z",
      "CL_100000": "spl_kB9lqsI4roUE",
      "CL_50000": "spl_jlYGh0C0E3HG",
      "CD_1000000": "spl_qYuy0H980Euh",
      "CD_500000": "spl_0tE5L6PgHtLu",
      "CD_200000": "spl_xwV6pCunH6I6",
      "CD_100000": "spl_1hWaK9N0ju3M",
      "CD_50000": "spl_LpjgXkzt9Xg",
    }
  }
}
```

```
class Airfoil(airfoil_data):
    def eval_cl(alpha, Re)
    def eval_cd(alpha, Re)
    def cl_to_cd(alpha, Re)
    def cd_to_cl(alpha, Re)
    def alpha_optimal(Re)
    def alpha_max_lift(Re)
    def alpha_min_drag(Re)
    ...
```

Example: Extracting and Reusing Historical Data

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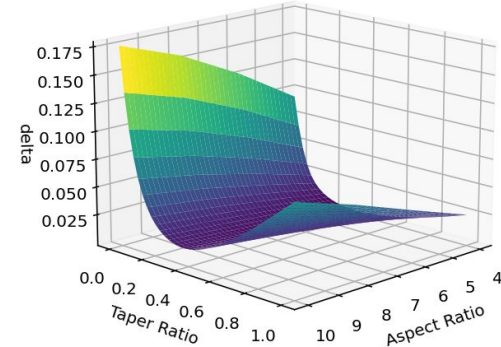
Python

```
delta_curves = {
    "AR=10": load_spline("spl_Nb5m075aHe42"),
    "AR=8": load_spline("spl_YUtYmXHXK5Q"),
    "AR=6": load_spline("spl_VsE31rlqsGtH"),
    "AR=4": load_spline("spl_lvdZRtwoGU6T")
}
```

Span Efficiency(AR, TR)

Induced Drag Coefficient

$$C_{Di} = \frac{C_L^2}{\pi AR} \cdot (1 + \delta)$$



https://splinecloud.com/compose/dfl_YRvflsQYc80M/relations

McCormick, B. W. (1979). Aerodynamics, Aeronautics, and Flight Mechanics (p. 652). John Wiley & Sons. <https://rexresearch1.com/FlightMechanicsLibrary/AerodynamicAeronauticsFlightMechanics.pdf>



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